MIL-P-63226 (AR)
AMENDMENT 3
21 August 1987
SUPERSEDING
AMENDMENT 2
21 June 1985

MILITARY SPECIFICATION

PROJECTILE, GUIDED,
155MM, XM712, WARHEAD FOR

This amendment forms a part of Military Specification MIL-P-63226 (AR), dated 1 August 1979, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "M712" to title and wherever it appears in this specification.

* SPECIFICATIONS, MILITARY, Add the following specifications:

"MIL-C-450C - Coating Compound Bituminous Solvent Type
MIL-I-6866 - Inspection, Penetrant Method of
MIL-H-6875 - Heat Treatment of Steels (Aerospace Practice, Process For)
MIL-I-8950 - Inspection, Ultrasonics, Wrought Metals, Process For
MIL-I-25135 - Inspection Materials, Penetrant
MIL-C-45662 - Calibration Systems Requirements"

PAGE 2

* DRAWINGS: Delete the following:

"9300922 - Adhesive, Explosive Compatible Epoxy, Two Component
9307009 - Seal, End Closure
9307046 - Wiring Harness, Warhead - 3W1 - 3W2
9300967 - Fabrication of Control Section and Warhead Body Housing, Process Specification For"

FSC

AMSC N/A

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DRAWINGS: Add the following:

"9300374 - Filler, Molded
9300786 - Adhesive, Explosive Compatible
9300790 - Adhesive, Epoxy, Two Component
9300792 - Powder Fluorescent
9300793 - Release Agent
9307062 - Seal, Cable Exit
9307069 - Wiring Harness"

Delete the following in its entirety:

"Equipment Operating Instructions

EOI 9304456 - Warhead Test Set, Equipment Operation Instruction"

* Add Para. 2.2 as follows:

"2.2 Other publications. The following documents form a part of this specification to the extent specified herein, unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8 - Tension Testing of Metallic Materials
E18 - Tests for Rockwell Hardness of Metallic Materials.
A370 - Standard Methods and Definitions for Mechanical Testing of Steel Products

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

ANSI-B46.1-1962 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)"

PAGE 3

3.2.1.1, line 8: Delete "9307046 and substitute "9307069".

3.2.1.2, line 8: Delete "9307046" and substitute "9307069".

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Delete the following: "This requirement does not include fuze wiring.".

3.2.1.3, line 2: Delete "installation" and substitute "electrical connection".

Line 3: Add "The electrical checks per Table 1 are not required, at the warhead section level, if performed as part of the warhead section/control section cable resistance checks."

PAGE 4

3.2.1.4, line 2: Delete "10^{-4}" and substitute "10^{-2}".

PAGE 6

3.2.1.5.7.2(b): Delete in its entirety and substitute the following:

"3.2.1.5.7.2(b) Cavitation, cracks or annular rings in segments A or B in excess of that permitted by the applicable requirement, but not in excess of twice that permitted, except that such cavities when located more than 1.00 inch from the warhead axis in segment A shal be disregarded."

PAGE 7

Add the following paragraphs:

"3.2.1.9 Bonding aft harness.

3.2.1.9.1 After bonding the harness to 4.03 diameter surface, assure smooth, void and air bubble free adhesive in slots and under and around cables.

3.2.1.9.2 After bonding glass cloth tape strips to 4.03 diameter, assure smooth continuous outer surface of adhesive and smooth continuous fillets covering all edges of the glass cloth tape. No voids, bubbles, or adhesive starved areas shall be visible.

3.2.1.9.3 After bonding the metal shims assure no voids, discontinuities of air bubbles are visible along any edge of shim.

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PAGE 8

* "3.5 Welding of warhead (if applicable). The weld zone, after heat treatment, shall meet the strength requirements of the finished item.

* 3.6 Warhead integrity.

3.6.1 Ultrasonic examination (if applicable). The weld zone shall be ultrasonically inspected using an angle beam method. No discontinuity shall be acceptable which is more than 1 DB below that obtained from the least responsive notch in the reference standard per Figure 3.

* 3.6.2 Fluorescent penetrant examinations. There shall be no indications of cracks, laps, seams, or tears when the fully machined part is inspected by fluorescent penetrant inspection.

* 3.7 Mechanical Properties. Mechanical properties of the Warhead Body shall be as required on the applicable drawing."

Add Figure 3 as it appears on the following page.

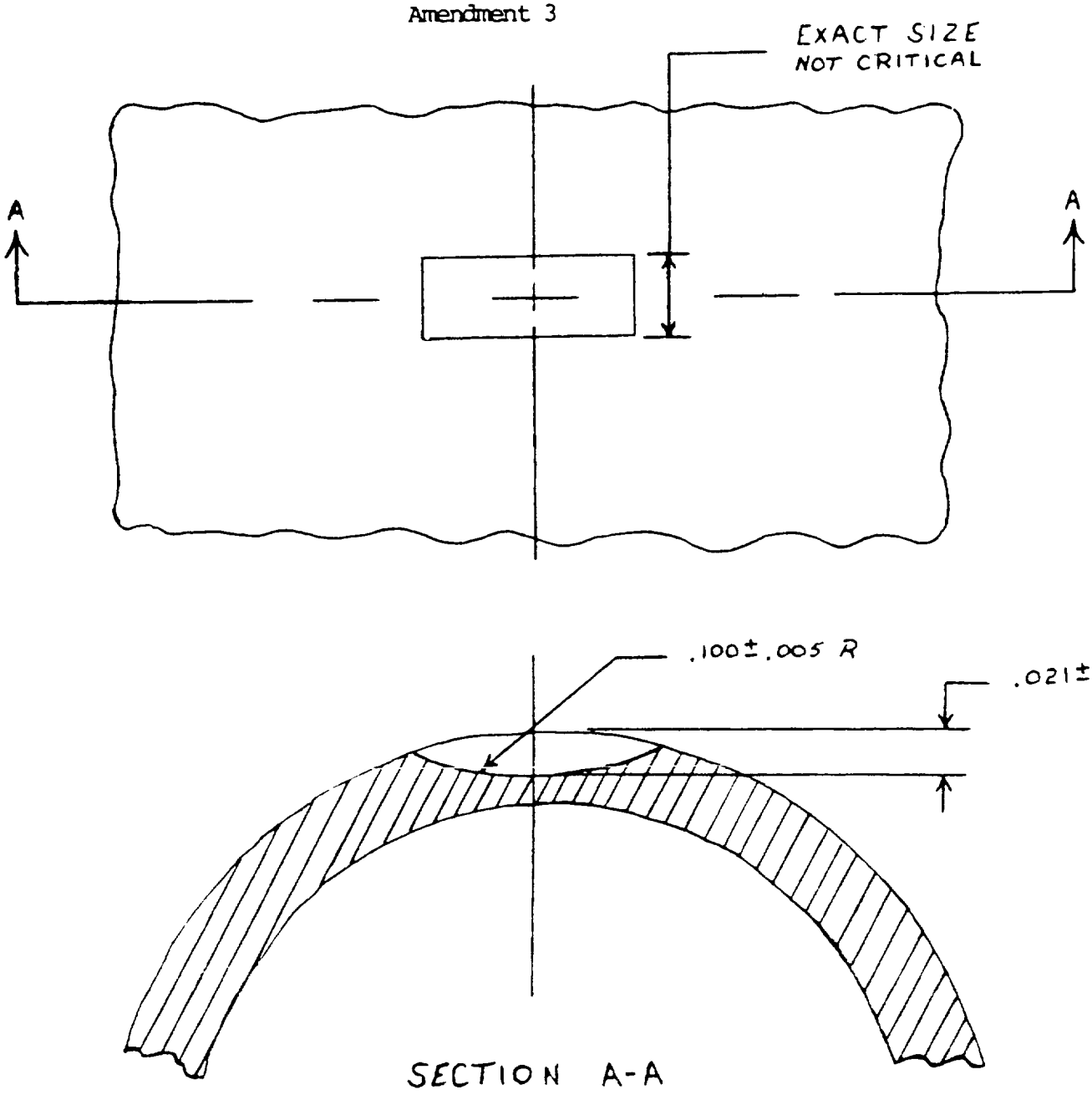
PAGE 9

Delete Paragraph 4.4.1 in its entirety and substitute the following new paragraph:

* "4.4.1 Inspection lot formation. Inspection lot size, lot formation, and presentations of lots shall be accordance with the inspection and sampling procedures herein and the provisions of MIL-STD-105. Warhead section lot sizes shall be no greater than 1 month's production. Each lot of warhead sections shall consist of: warhead forgings made from a single heat of steel; tubes made from a single heat of steel; fuze modules from not more than one interfix number; and Composition B which shall be of the same grade from one interfix, each from not more than one manufacturer.

* 4.4.1.1 Weld lot formation (if applicable). Warhead body weld lots shall not exceed one mill heat of steel for each component (forging and body), 400 components or the production yield of ten (10) consecutive working days, whichever is smallest. Additionally, weld lots shall be restricted to components welded and stress relieved in the same equipment, same process, with the same pressure, the same RPM and the same flywheel mass. The contractor may submit an alternate procedure for weld lot formation and inspection of weld lots, to the procuring activity for approval.

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NOTES:

1. NOTCH CROSS SECTIONAL AREA 0.00174 SQ. IN.
2. I.D. NOTCH HAS SAME DIMENSION AS SHOWN ABOVE

FIGURE 3

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* 4.4.1.2 Heat treatment lot formation. Heat treatment lots shall be defined as warhead bodies heat treated in one continuous run (see 4.4.1.2.1), and shall not exceed one mill heat of steel for each component (forging and body) or 300 components or 24 hours production, whichever is smallest. Additionally, heat treatment lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time, and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into large heat treatment lots after qualifying with individual batches, provided qualifying procedures are submitted to and approved by the Government and provided all the other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment lots shall remain grouped together and identified by lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots. The contractor may submit an alternate procedure for heat treatment lot formation and inspection of the heat treatment lot to the procuring acitivity for approval.

* 4.4.1.2.1 Continuous run. A continuous run is defined to be a run which involves only one part (i.e. control housing, warhead body, or electronic housing), with no interruption of the scheduled flow of parts into or out of the system, and no changes in the parameters of any furnace or bath.

4.4.2.2: Delete in its entirety and substitute the following:

"4.4.2.2 Lot acceptance. Lot acceptance shall be based on acceptance of individual sublots consisting of warheads poured from explosive melted at one time in a single melt kettle. Beginning with the first sublot produced and continuing until five consecutive sublots have met the requirements of drawing 9307058 (Confidential), two units shall be selected at random from each sublot for static penetration in accordance with 4.5.8. After five consecutive sublots have met the requirements of drawing 9307058 (Confidential), one unit shall be selected at random from each sublot for static penetration in accordance with 4.5.8. Failure of any loaded warhead assembly to meet the requirements of drawing 9307058 (Confidential) shall be cause for rejection of

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the sublot and processing in accordance with 4.4.3.2. Sublot acceptance tests shall be performed in accordance with Table VI."

4.4.3.2: Delete in its entirety and substitute the following:

"4.4.3.2 Sublot rejection. Any sublot failing to meet the requirements of 4.4.2.2 shall be resubmitted once by selecting two additional loaded warhead assemblies for test. This test set-up shall make provisions for flash x-ray to record radiographs of the shaped charge jet in two planes at 90 degrees apart. If both loaded warhead assemblies meet the requirements of drawing 9307058 (Confidential), the sublot shall be accepted. Failure to meet the requirements of drawing 9307058 (Confidential) shall cause the sublot to be rejected and the provisions of 4.4.2.2 shall again apply until five consecutive sublots are accepted. Rejected sublots, with test data and radiographs, shall be submitted to the procuring activity for disposition."

Add Paragraph 4.4.4 as follows:

* "4.4.4 Testing

* 4.4.4.1 Mechanical properties of warhead body.

* 4.4.4.1.1 Hardness. Each warhead body shall be Rockwell hardness tested per paragraph 4.5.13. These values will also be used for selection of tension testing.

* 4.4.4.1.2 Tensile test. Two warhead bodies, from each heat treatment lot shall be selected for this test (one body with the highest hardness reading and one with the lowest hardness reading). From each of these bodies, specimens shall be tested per paragraph 4.5.13 to determine compliance with ultimate tensile strength, yield strength and elongation requirements. Failure of any tensile specimens shall be cause for rejection of the heat treat lot.

* 4.4.4.1.2.1 Retest. If the high hardness body fails the yield strength requirement, the entire heat treatment inspection lot shall be rejected. In the event the lowest hardness body fails to meet the yield strength requirement, higher hardness values may be selected and tested in accordance with 4.4.4.1.2 in order to establish the acceptable minimum hardness. Once this minimum hardness has been established, all bodies with hardness value below the new minimum shall be rejected. If the low hardness body fails the elongation requirement, the entire heat treatment inspection lot shall

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be rejected. In the event the highest hardness body fails to meet the elongation requirement, lower hardness values may be selected and tested in accordance with 4.4.4.1.2 in order to establish the acceptable maximum hardness. Once this maximum hardness has been established, all bodies with hardness values above the new maximum shall be rejected.

* 4.4.4.2 Mechanical properties of warhead body weld (if applicable).

4.4.4.2.1 Tensile testing. The first and last body from each weld lot shall be selected for this test. Two (2) tensile bars shall be selected from each weld per paragraph 4.5.15 to determine compliance with ultimate tensile strength, yield strength and elongation requirements. Failure to meet the requirements of 3.5 shall be cause for rejection."

PAGE 10

4.5.3: Delete in its entirety and substitute the following:

"4.5.3 Leak test. The empty warhead assembly shall be pressurized to 15 $\pm$ 1.0 psig. The leakage rate shall not exceed the equivalent of 10^{-2} cc/sec of air at standard conditions. A leak rate of 10^{-4} cc/sec of air at standard conditions may be detected by immersion in water or other clear liquid for 30 seconds minimum. A continuous stream of bubbles will be emitted if the leak exceeds 10^{-2} cc/sec."

4.5.4.3: Delete in its entirety and substitute the following:

"4.5.4.3 Warhead section test. The electrical characteristics of the warhead section shall be tested and shall meet the requirements of Table I. If the fuze module was mechanically installed in the warhead prior to the time of the tests of 3.2.1.2 were conducted, it is not necessary to retest conductors 6 thru 18 in Table I. Resistance measurements shall be made with the positive output of the meter connected to the "From" conductor and the negative output of the meter connected to the "To" conductor of Table I. The test current shall be limited to 1 milliamp or less. The electrical checks per Table 1 are not required, at the warhead section level, if performed as part of the warhead section/control section cable resistance checks. However, if the warhead is supplied as GFM, then all of the tests in Table I are applicable."

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Add the following paragraph:

"4.5.12 Explosive leakage inspection. The warhead shall be inspected to verify conformance to the non-leakage requirement of 3.2.1.6."

Add paragraphs 4.5.13 through 4.5.17:

* "4.5.13 Hardness. In order to determine uniformity of heat treatment, each body after final heat treatment shall be subjected to a hardness test with a Rockwell Hardness Tester or any other method approved by the procuring activity. The test shall be made using the procedures for performing this test as prescribed by ASTM E-18 and ASTM A-370.

* 4.5.13.1 Heat treatment lots with warheads having forgings and bodies from the same mill heat of steel or from single piece warheads. The readings shall be taken on the aft face of the warhead.

* 4.5.13.2 Heat treatment lots with warheads having forgings and bodies from different mill heats of steel. The readings shall be taken on both the forward and the aft faces of the warhead. Contractor may submit an alternate procedure for determining the compatibility of heat treatment of forgings and tubing. If the alternate procedure is approved by the Government, then the dual hardness tests are not required and the readings shall be taken on the aft face of the warhead.

* 4.5.14 Tensile strength, yield strength and percent elongation of warhead body. The criteria for performing the tests shall be as prescribed by ASTM E-8.

* 4.5.14.1 Heat treatment lots of single piece warheads. The hardest and softest bodies as determined by the hardness test in 4.5.13.1 from each heat treat lot shall be subjected to this test. Four (4) tensile test specimens shall be selected, approximately 90° apart and shall be the largest obtainable specimen.

* 4.5.14.2 Heat treatment lots with warheads having the forgings and bodies from the same mill heat of steel. The hardest and softest warhead bodies as determined by the hardness test in 4.5.13.1 from each heat treat lot shall be subjected to this test. Four (4) tensile test specimens

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shall be selected, two from the body and two from the forging, approximately 180° apart and shall be the largest obtainable specimen as prescribed by ASTM E-8, Fig. 8 for the forging and Fig. 13 for the body.

* 4.5.14.3 Heat treatment lots with warheads having forgings and bodies from different mill heats of steel. The hardest and softest warhead forging section as determined by the hardness test in 4.5.13.2 from each heat treatment lot shall be subjected to this test. Four (4) tensile test specimens shall be selected from the forging section of the warhead, approximately 90° apart and shall be the largest obtainable longitudinal specimen as prescribed by ASTM E8, Fig. 8. Also, the hardest and softest warhead body section shall be subjected to this test. Four (4) tensile test specimens shall be selected from the warhead body section, approximately 90° apart and shall be a longitudinal specimen as prescribed by ASTM E-8, Fig. 13.

* 4.5.14.4 Elongation. Acceptance shall be based on the average elongation of all tensile specimens selected from each housing. If a tensile specimen fails within the weld zone, then that particular elongation value shall not be used in computing the average elongation. If the average elongation is less than 10 percent, then the lot represented by this test sample shall be rejected.

* 4.5.15 Tensile testing of weld. Four (4) tensile specimens shall be selected, 90° apart from the welds section per 4.4.4.2.1. The specimens shall be the largest obtainable specimen as prescribed by ASTM E-8. The weld cross-section shall be in the middle half of the gage length of the tensile specimen. The samples shall be taken after welding and stress relieving. The samples shall be heat treated to the requirements of the warhead body and tested per ASTM E-8. Failure to meet the applicable strength and elongation requirements shall be cause for rejection of the lot.

* 4.5.16 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be performed on 100 percent of the items after final machining and heat treating. Inspection shall be in accordance with Group VI, MIL-I-25135 and MIL-I-6866. Both sides of the part shall be examined. Parts examined by penetrant and found acceptable shall be ink stamped or stenciled with the letter "P". The marking shall withstand normal handling.

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* 4.5.17 Ultrasonic inspection (if applicable). Each lot of items shall be ultrasonically examined after weld flash removal and stress relief.

* 4.5.17.1 Procedure. Ultrasonic testing shall be conducted following the general principles outlined in MIL-I-8950. Calibration of the ultrasonic system using reference standards containing simulated discontinuities shall be performed at the beginning of each test run of given part, after each 30 minutes during the run, at the end of each run, and any time that a malfunction is suspected. If any check reveals that the system is not functioning properly, all of the material that has been examined since the last acceptable check shall be reexamined. Calibration shall be performed by obtaining response from each of the simulated discontinuities associated with the zone under test, setting the various controls such that the least of the responses triggers the alarm, and then increasing the system gain by 1 DB for production art testing.

* 4.5.17.2 Ultrasonic inspection standards. Reference standards shall be of the same thickness, material and configuration as the parts being examined. Simulated discontinuities will be circular segment notches circumferentially cut into the I.D. and O.D. of each weld zone. See Figure 3. For each weld zone I.D. and O.D. there will be 3 notches. One at the near side of the weld zone, one at their nominal center of the weld zone, and one in the far side of the weld zone. The reference standard shall be appropriately treated to inhibit corrosion."

PAGE 19

Add paragraph 6.5 and 6.6 as follows:

* "6.5 Weld upset (advisory). It has been found that acceptable welds have been produced with upset lengths of between 0.230 to 0.305 inches.

* 6.6 Holding grip marks (Advisory). It is recommended that holding grips be so configured as to not allow slippage of components not more than 0.100 inch. Furthermore, adequate surface should be provided to avoid the slippage, and to resist the torque developed during welding, without deforming the part."

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PAGE 22

Critical 1, 2, and 3: Delete in their entirety and substitute the following:

"1	No explosive material, other than within charge cavity	100%	3.2.1.6	4.5.12
2	Warhead cavity finish/ cleanliness	100%	3.2.1.7	4.5.9
3	Defects in explosive charge	100%	3.2.1.5.7.1	4.5.5 4.5.6"

PAGE 25

Major 108: Delete in it's entirety and substitute the following:

"108	Distance between face of liner and datums A and B	0.40%	Dwg	Gage"

PAGE 28

Critical: Delete the following in their entirety:

"1	Thickness	100%	Dwg.	Gage
2	Material	N/A	Dwg.	Vendor Lot Certification
3	Hardness	100%	Dwg.	Gage"

Major: Add the following:

"101	Thickness	100%	Dwg.	Gage
102	Material	NA	Dwg.	Vendor Lot Certification
103	Hardness	100%	Dwg.	Gage"

PAGE 33

Delete in it's entirety and substitute the following page:

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TABLE V
CONFORMANCE INSPECTIONS, SAMPLING

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9307001
	Body, Warhead				NEXT HIGHER ASSEMBLY 9307011
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Mechanical properties		4.4.4.1.2	3.7	4.5.14
2	Penetrant inspection for cracks		100%	3.6.2	4.5.16
3	Weld zone, yield and ultimate strength		4.4.4.2.1	3.5	4.5.15
Major					
101	Runout of forward face to forward and aft diameter datums		0.40%	Dwg.	Gage
102	Runout of ID in central portion of body to forward and aft diameter datums		0.40%	Dwg.	Gage
103	Runout of OD at aft end of maximum diameter to forward and aft diameter datums		0.40%	Dwg.	Gage
104	Runout of aft face to forward and aft diameter datums		0.40%	Dwg.	Gage
105	True position of fuze well diameter		0.40%	Dwg.	Gage
106	True position of thread at forward end		0.40%	Dwg.	Gage
107	Liner centering diameter		0.40%	Dwg.	Gage

NOTES:

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Delete in it's entiretv and substitute the following page:

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TABLE V
CONFORMANCE INSPECTIONS, SAMPLING

CLASSIFICATION OF DEFECTS & TESTS

		SHEET 1 OF 1		DRAWING NUMBER 9307001
				NEXT HIGHER ASSEMBLY 9307011
OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
er centering		0.40%	Dwg.	Gage
D				
ting shoulder to		0.40%	Dwg.	Gage
		0.40%	Dwg.	Gage
l		100%	Dwg	Gage
		100%	Dwg.	Visual
urface				
ace shoulder to		100%	Dwg.	Gage
tums		100%	Dwg.	Gage
ace cylinder				
g groove from		100%	Dwg.	Gage
lder		4.4.4.2.1	3.5	4.5.15
n		2.5%	3.6.1	4.5.17
n of weld		100%	N/A	4.5.13
		0.65%	Dwg.	Visual
		0.65%	Dwg.	Gage
able notch		0.65%	Dwg.	Gage
oles from forward		0.65%	Dwg.	Gage
erface shoulder		0.65%	Dwg.	Gage
		0.65%	Dwg.	Gage

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PAGE 38

Critical: Delete the following in their entirety:

"1	No unsolified paint in charge cavity	100%	3.2.1.7	4.5.9
2	Finished surface	100%	3.2.1.7	4.5.9
3	Adhesive	N/A	Spec.	9300922 - Vendor lot Certification"

Critical: Add the following:

"1	Adhesive explosive compatibility	N/A	Dwg.	9300786
2	Bond of aft harness	100%	3.2.1.9.1	Visual
3	Bond of glass cloth	100%	3.2.1.9.2	Visual
4	Bond of metal shim	100%	3.2.1.9.3	Visual
5	Adhesive	N/A	Spec.	9300790 - Vendor lot Certification"

PAGE 39

Critical 1, 2, and 3: Delete in their entirety and substitute the following:

"1	No explosive material other than within charge cavity	100%	3.2.1.6	4.5.12
2	Warhead cavity finish/ cleanliness	100%	3.2.1.7	4.5.9
3	Defects in explosive charge	100%	3.2.1.5.7.1	4.5.5 4.5.6 Review existing radio-graphs"

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"The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodians:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A977)

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